

Higher-Order Metaphysics
Week Seven: Classicism and questions of grain
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1 Classicism

Recall that we defined Classicism to be the result of closing H under the

Rule of Substitution If $\vdash P \leftrightarrow Q$ and $\vdash R[P/p]$ then $\vdash R[Q/p]$.

It can also be characterized as the result of closing H under either of the following rules:

Rule of Equivalence (version 1) If $\vdash P \leftrightarrow Q$, $\vdash (\lambda \vec{x}.P) = (\lambda \vec{x}.Q)$.

Rule of Equivalence (version 2) If $\vdash R\vec{x} \leftrightarrow S\vec{x}$, $\vdash R = S$.

or as the result of adding either of the following axiom-schemes to H:

Logical Equivalence (version 1) $(\lambda \vec{x}.P) = (\lambda \vec{x}.Q)$, where $\vdash_H P \leftrightarrow Q$.

Logical Equivalence (version 2) $R = S$, where $\vdash_H R\vec{x} \leftrightarrow S\vec{x}$,

or as the result of adding a certain list of closed identities to H.

The paper ‘Classicism’ uses a type system where e is not a terminal type. Confusingly, Bacon ends up defining ‘Classicism’ in such a way that in his standard type system (where e is a terminal type), it includes all instances of a certain “Modalized Functionality” axiom, including instances for complex types ending in e , which do not follow from the Rule of Substitution.

2 Some weakenings of Classicism

Booleanism restricts Logical Equivalence to the case where the biconditional is a theorem of *propositional* logic.

Quasi-Classicism restricts Logical Equivalence to the case where the biconditional is a theorem of *free* H—the logic like H except that it gets rid of UI (or $\forall$ Elim) and instead has the following axioms:

(QK) $(\forall v.P \rightarrow Q) \rightarrow (\forall v.P) \rightarrow (\forall v.Q)$

(QD) $(\forall v.P \rightarrow Q) \leftrightarrow (P \rightarrow (\forall v.Q))$ (v not free in P)

L-Classicism: is the result of closing H under the *rule of L-equivalence*: if $\vdash P \leftrightarrow Q$, $\vdash (\lambda \vec{x}.P \wedge Q) = (\lambda \vec{x}.P \vee Q)$.

3 Classicism and modal logic

In Classicism, the operator $\Box_{\top}$ satisfies the modal logic S4:

$$K \vdash \Box_{\top}(P \rightarrow Q) \rightarrow \Box_{\top}P \rightarrow \Box_{\top}Q$$

$$T \vdash \Box_{\top}P \rightarrow P$$

$$4 \vdash \Box_{\top}P \rightarrow \Box_{\top}\Box_{\top}P$$

Necessitation If $\vdash P$ then $\vdash \Box_{\top}P$.

Some noteworthy theorems about $\Box_{\top}$:

- $\Box_{\top}(\forall \vec{x}.(R\vec{x} \leftrightarrow S\vec{x})) \rightarrow R = S$ (Intensionalism)
- $\Box_{\top}\forall_{\sigma_1}x.(Fx =_{\sigma_2 \rightarrow \dots \rightarrow \sigma_n \rightarrow t} Gx) \rightarrow F =_{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow \tau} G$ (Modalized Functionality)
- $x = y \rightarrow \Box_{\top}(x = y)$ (NI)
- $\Box_{\top}(\forall x.P) \rightarrow \forall x.\Box_{\top}P$ (CBF)
- $\forall x.\Box_{\top}\exists y.y = x$ (Broad Necessitism)
- $N(p \rightarrow p) \rightarrow \forall q(\Box_{\top}q \rightarrow Nq)$
- $N(p \rightarrow p) \rightarrow \forall x.N(\exists y.y = x)$

Some noteworthy non-theorems about $\Box_{\top}$:

- $P \rightarrow \Box_{\top}\Diamond_{\top}P$ (B)
- $\Diamond_{\top}P \rightarrow \Box_{\top}\Diamond_{\top}P$ (5)
- $x \neq y \rightarrow \Box_{\top}(x \neq y)$ (ND)
- $(\forall x.\Box_{\top}P) \rightarrow \Box_{\top}(\forall x.P)$ (BF)

4 Some strengthenings of Classicism

A natural direction to explore for strengthening is to add some or all of the above non-theorem schemes, or their necessitations, to Classicism.¹

In fact, B, 5, and ND are all *equivalent* in Classicism.

BF, meanwhile, is equivalent to the following principle, which we have met before:²

$$(\text{Functionality}) \quad \forall_{\sigma_1}x.(Fx =_{\sigma_2 \rightarrow \dots \rightarrow \sigma_n \rightarrow t} Gx) \rightarrow F =_{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow \tau} G$$

- $\Box_{\top}B$ (or $\Box_{\top}5$ or $\Box_{\top}ND$) implies $\Box_{\top}BF$ (and thus $\Box_{\top}\text{Functionality}$).
- B (or 5 or ND) and BF (or Functionality) jointly imply $\Box_{\top}B$ (and thus $\Box_{\top}5$, $\Box_{\top}ND$, $\Box_{\top}BF$, $\Box_{\top}\text{Functionality}$).

We call the result of adding $\Box_{\top}B$ to Classicism C5.

¹ Note that if we just add them as axioms, the resulting theory will no longer be closed under the Rule of Substitution or Necessitation, and we won't get the necessitated versions.

² BF does not imply Functionality for types ending in e .

5 Assuming Classicism, is $\Box_{\top}$ the broadest necessity?

We understand ‘being a necessity’ as of type $(t \rightarrow t) \rightarrow t$ (a property of properties of propositions), and ‘being at least as broad as’ as of type $(t \rightarrow t) \rightarrow (t \rightarrow t) \rightarrow t$ (a relation between properties of propositions).

Bacon’s strategy: first argue for specific analyses of these notions. Then show that under these analyses, Classicism implies that $\Box_{\top}$ is a necessity that is at least as broad as every necessity, and such that no distinct necessity is at least as broad as it.

He first defines being a *weak* necessity: roughly, X is a weak necessity iff every theorem of the modal logic K is true when $\Box$ is interpreted as X and the proposition letters are interpreted arbitrarily. But he rejects this as a definition of “necessity operator” on the grounds that *being believed by a* need not be a necessity operator even when a is *in fact* logically omniscient and perfectly introspective.

Instead, he defines ‘ X is a necessity operator’ as ‘For every weak necessity Y , $Y(X \text{ is a weak necessity})$ ’.

But this might be thought too demanding: mightn’t there be a logically omniscient and perfectly introspective agent who doesn’t believe that—say—physical necessity is a weak necessity?

Note however that to argue that $\Box_{\top}$ is a necessity assuming Classicism, we only need that being a necessity in Bacon’s sense is *sufficient* for being a necessity.³

³ Dorr, Hawthorne, and Yli-Vakkuri (2021: Appendix D) pursue a different strategy, which does not require committing to any specific analysis of ‘necessity operation’ or ‘broader than’, but just consider separate sufficient and necessary conditions.

6 Actuality and ND

Williamson (1996) gives the following argument for the necessity of distinctness, in a language with operators @ for ‘actually’ and $\Box$ for ‘necessarily’:

- | | |
|--|---------|
| 1. $\forall xy. \Box(x = y \rightarrow @(x = x) \rightarrow @(x = y))$ | Premise |
| 2. $\forall x. \Box @(x = x)$ | Premise |
| 3. $\forall xy. \Box (\neg @(x = y) \rightarrow x \neq y)$ | 1, 2 |
| 4. $\forall xy. \Box (@(x \neq y) \rightarrow \neg @(x = y))$ | Premise |
| 5. $\forall xy. (x \neq y \rightarrow \Box @(x \neq y))$ | Premise |
| 6. $\forall xy. (x \neq y \rightarrow \Box(x \neq y))$ | 3, 4, 5 |

There is some plausibility to the thought that every allowable interpretation of ‘necessarily’ in natural language fits with the interpretation of ‘actually’ (or “modal anaphora”) in such a way that this argument goes through.

That doesn't look strong as an argument for C5. It looks better as an argument that if ND fails, $\Box_{\top}$ is not an allowable interpretation of 'necessarily' in natural language.

Dorr, Hawthorne, and Yli-Vakkuri (2021: ch. 8) discuss a companion argument for BF:

- | | |
|---|---------|
| 1. $(\forall x.\Box Fx) \rightarrow (\Box @ \forall x.\Box Fx)$ | Premise |
| 2. $\Box (@(\forall x.\Box Fx) \rightarrow (\forall y.@\Box Fy))$ | Premise |
| 3. $(\Box \forall y.@\Box Fy) \rightarrow (\Box \forall y.Fy)$ | Premise |
| 4. $(\forall x.\Box Fx) \rightarrow (\Box \forall y.Fy)$ | 1, 2, 3 |

7 Two other necessity-like operators we can define in Classicism

$$\Box_{\neq} := \lambda p.\exists q.q \wedge \Box_{\top}(\Diamond_{\top}q \rightarrow p)$$

$$\Box_5 := \lambda p.\exists q.q \wedge \Box_{\top}(\Diamond_{\neq}q \rightarrow p)$$

Classicism proves that ND holds for $\Box_{\neq}$ and that 5 and B hold for $\Box_5$.⁴ C5 proves that they are all the same, but C doesn't.

⁴ Dorr, Hawthorne, and Yli-Vakkuri 2021: Appendix D.

Depending on how much logical well-behavedness one builds into "being a necessity operation", this suggests that $\Box_{\neq}$ or $\Box_5$ might be better deservers of the name 'the broadest necessity'.

8 Assuming Classicism, is $\Box_{\top}$ metaphysical necessity?

Kripke's characterization of his topic: what is 'necessary in the highest degree—whatever that means'.

9 "At least as coarse-grained as"

Intuitively, Classicism is a rather "coarse-grained" theory of higher-order reality, though not as coarse-grained as Extensionalism.

Bacon and Dorr (forthcoming: 2.4) suggest the following operational definition: T_1 is at least as coarse-grained as T_2 iff (i) every closed identity in T_2 is in T_1 , and (ii) every closed distinctness claim in T_1 is in T_2 .⁵

This suggests an interesting question: which theories have a maximally coarse grained consistent strengthening? Which have a maximally fine grained consistent strengthening?

It turns out that H_0 , H, and C all have consistent maximally fine grained strengthenings: the result of adding all distinctness claims consistent to the theory to the theory.

⁵ This seems too focused on closed as opposed to quantified identities and distinctness claims to be entirely defensible, but it at least gives something interesting to investigate.

References

- Bacon, Andrew, and Dorr, Cian (forthcoming), "Classicism", in Peter Fritz and Nicholas K. Jones (eds.), *Higher-Order Metaphysics* (Oxford: Oxford University Press).
- Dorr, Cian, Hawthorne, John, and Yli-Vakkuri, Juhani (2021), *The Bounds of Possibility: Puzzles of Modal Variation* (Oxford: Oxford University Press).
- Williamson, Timothy (1996), "The Necessity and Determinacy of Distinctness", in Savina Lovibond and Stephen G. Williams (eds.), *Essays for David Wiggins: Identity, Truth and Value* (Oxford: Blackwell), 1–17.